Oracle 4 Node RAC ASM Disk Addition Runbook

Step by step procedure for preparing 10 shared disks using fdisk partitioning and adding the partitions to ASM through the existing ASM management UI.

Assumption: Grid Infrastructure and RDBMS software are already installed on a 4 node RAC. The new disks are shared between all four nodes and the existing environment uses partitioned disks for ASM.

> **Important:** Do not execute destructive commands until disk identity, size, serial number, and shared storage presentation have been verified on all four nodes.

1. Overall Procedure

Step	Activity	Purpose
1	Identify the 10 new disks	Confirm disk identity before modifying anything.
2	Check existing partitions and signatures	Ensure the disks are not already in use.
3	Verify shared visibility on all 4 nodes	RAC ASM requires all nodes to see the same shared LUNs.
4	Partition disks with fdisk	Create one partition covering each disk.
5	Refresh kernel partition information	Ensure Linux recognizes the new partitions.
6	Verify partitions on all 4 nodes	Confirm consistent shared storage visibility.
7	Check ASM disk discovery	Confirm ASM sees the new partitions as candidates.
8	Add disks using the existing ASM UI	Add the correct partition paths to the intended disk group.
9	Verify ASM membership and capacity	Confirm the disks became ASM members.
10	Validate from all 4 RAC nodes	Confirm the disk group is visible cluster wide.

2. Prechecks

2.1 Identify the disks

Run on Node 1:

```
lsblk -o NAME,KNAME,TYPE,SIZE,FSTYPE,MOUNTPOINTS,MODEL,SERIAL
```

Why: This identifies the candidate disks and provides size, filesystem, model, and serial information. Do not rely only on /dev/sdX names because device numbering is not a reliable storage identity.

2.2 Check existing partitions

For each candidate disk:

```
fdisk -l /dev/sdb fdisk -l /dev/sdc fdisk -l /dev/sdd
```

Why: This confirms whether the disks already contain partitions. If a disk already has a partition or appears to contain data, stop and investigate before modifying it.

2.3 Check existing signatures without changing anything

Run for each candidate disk:

```
wipefs -n /dev/sdb wipefs -n /dev/sdc wipefs -n /dev/sdd
```

Why: The -n option performs a read only check. It can reveal filesystem, LVM, ASM, or other signatures. Do not use wipefs without -n on uncertain disks.

2.4 Verify all 4 RAC nodes see the shared disks

Run on Node 1, Node 2, Node 3, and Node 4:

```
lsblk -o NAME,KNAME,TYPE,SIZE,FSTYPE,MOUNTPOINTS,MODEL,SERIAL
```

Why: ASM disk groups in RAC require shared storage. The same physical LUN must be visible to all four nodes. If one node cannot see a disk, resolve the storage presentation issue before adding it to ASM.

3. Partition the Disks Using fdisk

Example for the first disk:

```
fdisk /dev/sdb
```

Inside fdisk, create one primary partition using the complete disk:

```
n p 1 Enter Enter w
```

After this, the disk should appear as:

```
/dev/sdb ■■ /dev/sdb1
```

Why: The partition provides a defined storage boundary that can be presented to ASM. This follows the partition based ASM preparation method used in the existing environment.

3.1 Repeat for all 10 disks

Use the actual device names identified during the precheck. Example:

```
fdisk /dev/sdb fdisk /dev/sdc fdisk /dev/sdd fdisk /dev/sde fdisk /dev/sdf fdisk /dev/sdg fdisk /dev/sdh
fdisk /dev/sdi fdisk /dev/sdj fdisk /dev/sdk
```

For each disk, create partition 1 using the complete disk. Do not assume these are the correct disks without checking them first.

4. Verify Partition Creation

```
lsblk
```

Expected pattern:

```
sdb 500G ■■sdb1 500G sdc 500G ■■sdc1 500G
```

Why: This confirms that the partition table was written and Linux can see the new partition.

5. Refresh the Kernel Partition Table

Usually fdisk causes the kernel to recognize the partition immediately. If the new partition does not appear:

```
partprobe /dev/sdb
```

Then check:

```
lsblk
```

Why: partprobe asks the running kernel to reread the partition table without requiring a reboot.

6. Verify the Shared Partitions on All 4 Nodes

Run on all nodes:

```
lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINTS,MODEL,SERIAL
```

Each node should show the corresponding partition, for example /dev/sdb1, /dev/sdc1, and so on.

Why: Because the LUNs are shared, the partition created on the shared disk should become visible from all RAC nodes. Do not independently create different partitions on each node for the same shared LUN.

7. Check ASM Discovery

Switch to the Grid owner:

```
su - grid
```

Connect to ASM:

```
sqlplus / as sysasm
```

Check ASM disk discovery:

```
show parameter asm_diskstring;
```

Then:

```
select path, header_status, mode_status, state, total_mb from v$asm_disk order by path;
```

Why: ASM_DISKSTRING controls where ASM looks for disks. The new partition paths must be discoverable by ASM before they can be added to a disk group.

8. Check HEADER_STATUS Before Adding

For new disks, you normally expect CANDIDATE or, depending on the environment, PROVISIONED or FORMER.

HEADER_STATUS	Meaning / Action
CANDIDATE	Available for ASM. Normally suitable for adding.
PROVISIONED	Prepared for ASM. Verify environment and intended disk group.
MEMBER	Already belongs to an ASM disk group. Do not add elsewhere.
FORMER	Previously belonged to an ASM disk group. Verify it is safe to reuse.
FOREIGN	Contains non ASM metadata. Stop and investigate.
UNKNOWN	ASM cannot determine the disk state. Investigate before proceeding.

9. Add the Partitions Through the Existing ASM UI

Open the same ASM management UI used previously. Select the intended ASM disk group and add the newly discovered partition devices.

Important: Select the partition devices, for example /dev/sdb1, not the parent disks such as /dev/sdb, when the existing setup is partition based.

Conceptually, the ASM operation performed by the UI is similar to:

```
ALTER DISKGROUP DATA ADD DISK '/dev/sdb1', '/dev/sdc1', '/dev/sdd1';
```

Do not execute the example SQL blindly. Use the actual ASM disk group and actual discovered paths from your environment.

10. Verify ASM After Addition

```
asmcmd lsdg
```

Then:

```
select name, state, type, total_mb, free_mb, usable_file_mb from v$asm_diskgroup;
```

And verify individual disks:

```
select group_number, disk_number, name, path, header_status, state, total_mb from v$asm_disk order by
group_number, disk_number;
```

Expected: The newly added disks should show HEADER_STATUS as MEMBER and should belong to the intended disk group.

11. Final RAC Validation

From each RAC node, check:

```
asmcmd lsdg
```

From ASM SQL, validate all instances:

```
select inst_id, name, state, type, total_mb, free_mb from gv$asm_diskgroup order by inst_id, name;
```

Why: The disk group must be visible consistently through all four ASM instances. This confirms that the RAC cluster has a consistent view of the shared storage.

12. Final Validation Checklist

■ All 10 disks identified using size and persistent storage information.

■ No existing filesystem, LVM, or unexpected ASM signature found.

■ All 10 shared disks visible from Node 1, Node 2, Node 3, and Node 4.

■ One partition created on each disk using fdisk.

■ Partitions visible as /dev/sdX1 on all nodes.

■ ASM_DISKSTRING allows ASM to discover the partition paths.

■ New disks show an appropriate HEADER_STATUS before addition.

■ Correct partition paths selected in the ASM management UI.

■ New disks show HEADER_STATUS = MEMBER after addition.

■ ASM disk group capacity increased as expected.

■ ASM disk group is visible from all four RAC nodes.

13. Useful Command Reference

Disk inventory

```
lsblk -o NAME,KNAME,TYPE,SIZE,FSTYPE,MOUNTPOINTS,MODEL,SERIAL
```

Partition details

```
fdisk -l /dev/sdX
```

Read only signature check

```
wipefs -n /dev/sdX
```

Refresh partition table

```
partprobe /dev/sdX
```

ASM disk discovery

```
select path, header_status, mode_status, state, total_mb from v$asm_disk order by path;
```

ASM disk groups

```
select name, state, type, total_mb, free_mb, usable_file_mb from v$asm_diskgroup;
```

Operational caution: The most dangerous steps are disk identification and commands that write or remove storage metadata. Always validate the actual device, serial number, size, and shared storage mapping before running fdisk or any disk initialization command.